

LITTLE STREAM DETECTOR

LSD 3.0 Detailed Technical Documentation

Property	Value
Release	3.0
Documentation date	2 September 2026
Runtime	Windows PowerShell 5.1 and .NET Framework
External multimedia tools	None
Temporary media files	None
Validated VP9 route	Matroska/WebM V_VP9 and non-fragmented MP4 vp09, Profile 0, 8-bit YUV 4:2:0
Implemented VP9 route	Legacy MP4 vp08 mapping; separate runtime asset not yet tested
Known container limit	Fragmented MP4 using moof/traf/trun is not supported

Scope. This edition documents the stable LSD 3.0 architecture, native VP9 frame analysis, VP9 superframe traversal, frame-level Base Q Index histograms, and the shared container-neutral route used by Matroska/WebM and standard non-fragmented MP4/MOV.

Validation state. VP9 in Matroska/WebM is regression validated. The MP4/MOV sample-entry mapping is implemented through the same canonical VP9 parser and should be runtime-validated with a representative VP9-in-MP4 asset before the route is described as regression validated.

1. Purpose and System Scope

Little Stream Detector 3.0 is a native Windows PowerShell application with an embedded C# analysis engine. It detects supported media containers, builds bounded container-neutral samples, performs codec-aware bitstream analysis, and renders human-readable and structured results in a responsive graphical interface.

The VP9 extension adds deterministic traversal of VP9 uncompressed frame headers without changing container addressing, the canonical data model, or Current A / Reference B comparison semantics.

1.1 Primary objectives

- Perform native analysis without FFmpeg, MedialInfo, or other external multimedia processes.
- Keep container-specific addressing separate from codec-specific interpretation.
- Validate every canonical sample against file boundaries before reading payload data.
- Report unavailable data as N/A rather than guessing.
- Support asynchronous preparation and analysis without blocking the GUI.
- For VP9, distinguish container samples, internal coded frames, shown frame events, hidden coded frames, and show-existing events.

1.2 Supported families

Area	Supported families
Containers	Matroska, WebM, AVI 1.0, OpenDML AVI 2.0, MP4, M4V, MOV
Video	H.264/AVC, H.265/HEVC, AV1, VP9, MPEG-4 Part 2/XviD
Audio	AAC-LC, HE-AAC, HE-AACv2, MP3, PCM integer/float, AC-3, E-AC-3
Outputs	Summary A, optional Summary B, Streams, JSON, Log, bitrate profile, quantizer distribution

2. Architectural Overview

LSD is a modular monolith. PowerShell owns application state, presentation, orchestration, and background-job lifecycle. Embedded C# owns binary parsing, canonical models, bitstream traversal, validation, and numerical accumulation.

Pipeline. User/GUI -> metadata and job coordinator -> Matroska, MP4, or AVI reader -> canonical media model -> video and audio analyzers -> completed result snapshot -> Current A / Reference B comparison views.

Layer	Responsibility	Representative components
Presentation	Window, reports, graphs, drag-and-drop, active-tab copy	PowerShell WinForms
Coordination	Metadata preparation, jobs, cancellation, progress	Start-Meta, Complete-NativeMetadata, Start-Scan
Container parsing	Tracks and canonical sample locations	NativeMediaProbe, NativeMp4Probe, NativeAviProbe
Canonical adaptation	Normalize tracks and bounded samples	MatroskaToLsdAdapter, Mp4ToLsdAdapter, AviToLsdAdapter
Codec analysis	Configuration and frame-level syntax	AVC, HEVC, AV1, VP9, MPEG-4 Visual
Comparison	Retain B and render A/B views	ReferenceResult, ReferenceOnlyView

2.1 VP9 isolation

V_VP9 maps to CodecFamily VP9 and PayloadFormat Vp9Frame. Vp9FrameParser is selected only for VP9. The same parser receives canonical samples from validated Matroska/WebM input and from mapped standard non-fragmented MP4/MOV input. No container-specific VP9 parser is introduced.

3. Runtime and Processing Lifecycle

Phase	Action	Output
1. Selection	Select or drop a file	Absolute source path
2. Native detection	Identify container	NativeMediaInfo
3. Metadata preparation	Parse tracks and sample indexes	Container probe result
4. Canonical adaptation	Create common bounded samples	LsdMediaModel
5. Codec analysis	Parse frame and audio syntax	Diagnostics and statistics
6. Finalization	Build report and histograms	Current A result
7. Optional promotion	Set Ref B	Reference B snapshot
8. Comparison	Analyze later A	A/B views

3.1 VP9 sample lifecycle

Each bounded Vp9Frame sample is read once. A valid superframe index can split one sample into multiple internal VP9 frames. Internal frames are parsed in coded order so reference-slot geometry remains available. The sample contributes one bitrate observation, while every newly coded internal frame contributes one Base Q Index observation.

A show_existing_frame event contributes a display event but does not create a new Base Q Index observation.

4. Canonical Media Model

Object	Key fields or meaning
LsdMediaModel	SourcePath, ContainerType, FileLength, DurationSeconds, TimecodeScale, Tracks
LsdTrackModel	TrackId, TrackType, CodecId, CodecFamily, CodecPrivate, PayloadFormat, Samples
LsdSampleModel	FileOffset, Length, DTS, PTS, Duration, key/invisible/discardable flags, PayloadFormat

4.1 VP9 accounting semantics

A canonical VP9 sample is a container payload range, not necessarily one coded frame. A VP9 superframe may carry one shown frame plus one or more hidden coded frames. LSD reports Video samples, Internal VP9 frames, Shown frame events, Hidden coded frames, and Show existing frame separately.

4.2 Reference snapshot

Reference B is a presentation-layer snapshot containing file identity, duration, bitrate bins, quantizer mode, quantizer counts, and frozen Summary B text. VP9 uses the distinct quantizer mode VP9 Base Q Index.

5. Container Modules

5.1 Matroska and WebM

The Matroska reader maps V_VP9 to CodecFamily VP9 and PayloadFormat Vp9Frame. This route is regression validated.

5.2 MP4, M4V, and MOV

The ISO Base Media reader supports standard non-fragmented files with moov-based sample tables. The video sample-entry reader recognizes vp09 and legacy vp08, maps both entries to V_VP9, retains the optional vpcC child box, and routes bounded samples to the existing Vp9FrameParser.

- stsd identifies the video sample entry.
- stsz, stsc, and stco/co64 resolve sample locations.
- stts, cts, and stss provide timing and sync information.
- The same canonical VP9 analysis path is reused after sample construction.

5.3 AVI

VP9 in AVI is not enabled or validated. Existing AVI support remains unchanged for supported codec mappings.

5.4 Fragmented MP4

Fragmented MP4 using moof, traf, tfhd, tfdt, and trun is outside the current LSD 3.0 scope. Such files require a separate fragment-aware sample indexer before codec analysis can begin.

6. Video Analysis Modules

Family	Native analysis
H.264 / AVC	avcC and Annex B, SPS/PPS, I/P/B, SliceQPY, CABAC/CAVLC, VUI
H.265 / HEVC	hvcC, VPS/SPS/PPS, NAL class, first-slice I/P/B, SliceQPY, VUI
AV1	av1C, sequence/frame OBU traversal, hidden/show-existing, Base Q Index
VP9	Superframe index, uncompressed header, reference slots, shown/hidden events, KEY/INTER, Base Q Index
MPEG-4 Part 2 / XviD	VOL/VOP, I/P/B/S-VOP handling, packed samples, VOP quantizer

6.1 VP9 parser scope

- Validate the two-bit frame marker and profile signaling.
- Recognize show_existing_frame without inventing a new quantizer value.
- Parse KEY_FRAME, INTER_FRAME, and intra-only syntax needed to reach quantization.
- Track eight reference slots so frame_size_from_refs can resolve geometry.
- Split valid superframes using the trailing duplicated marker and little-endian frame sizes.
- Read base_q_idx and frame-level delta-Q fields.
- Classify displayed KEY frames as I and displayed INTER frames as P. VP9 has no classic B-frame class in the LSD report model.

6.2 Quantizer meaning

VP9 Base Q Index is a frame-level lookup-table index from 0 through 255. Segmentation may override quantization for individual blocks, so LSD does not describe the histogram as a reconstructed block-level quantizer distribution.

7. Audio Analysis Modules

Family	Validation and reporting
AAC	AudioSpecificConfig, core/output rate and channels, SBR, Parametric Stereo, profile naming
MP3	Version/layer, bitrate and sample-rate index, padding, frame size, decoded samples
AC-3 / E-AC-3	Syncword, bitstream ID, sample rate, frame size, blocks, channels, LFE
PCM	Block alignment, payload divisibility, decoded duration, expected and measured bitrate

Audio analysis remains independent of the VP9 extension. VP9 introduces no alternate audio path and does not change Summary B behavior.

8. Quantizer and Bitrate Analysis

8.1 Bitrate profile

Each canonical video sample contributes byte size to presentation-time bins. Internal hidden VP9 frames remain inside the containing sample and do not create synthetic timing or bitrate bins. Reference B is gold and Current A is blue.

Codec	Native value	Histogram unit
AVC	SliceQPY	Frame-level SliceQPY
HEVC	SliceQPY	Frame-level SliceQPY
AV1	Base Q Index	New coded frame
VP9	Base Q Index	New internal coded frame
MPEG-4 Part 2	VOP quantizer	Coded VOP

8.2 VP9 completeness

- A Base Q Index is emitted only for a newly coded VP9 frame.
- `show_existing_frame` produces a display event but no new Base Q Index.
- One superframe may emit multiple Base Q Index values from one canonical sample.
- The histogram is enabled only when the VP9 header set is complete and contains no rejected headers.
- Frame accounting and shown/sample consistency are reported explicitly.

8.3 Display behavior

The right panel uses the active quantizer mode in its title. VP9 is shown as VP9 Base Q Index distribution. Contextual histogram padding is constrained to the valid codec domain, so impossible negative or over-range labels are not displayed.

9. GUI, Jobs, Reporting, and Comparison

View	Content
Summary A	Readable report for Current A, including VP9 diagnostics
Summary B	Frozen reference report while B exists
Streams	Structured current native-track data
JSON	Serializable current metadata and streams
Log	Current lifecycle and validation messages
Bitrate graph	Blue A and gold B with role legend
Quantizer panel	Mode-aware A/B statistics, bars, percentages, and codec-specific title

9.1 VP9 report block

The Native VP9 Frame / Base Q Analysis block reports samples, internal frames, superframes, parsed and rejected headers, show-existing events, shown and hidden frames, KEY/INTER counts, Base Q statistics, first failure, and explicit validation states.

9.2 Copy action

Copy follows the active report tab. VP9 adds no alternate copy path and does not alter Summary B freeze behavior.

10. Reference Comparison State Model

The workflow supports exactly one Current A and one Reference B. It is not a queue and does not rank files.

State sequence. EMPTY -> open -> A blue -> Set Ref B -> B gold -> open -> B gold + A blue -> Remove Ref B -> A blue, or EMPTY when B was the only result.

- Every newly opened or dropped file becomes Current A and is blue.
- Set Ref B promotes completed A to B and waits for a new A.
- Opening another file while B exists preserves B and creates a new blue A.
- Remove Ref B removes only B.
- B is never silently converted back into A.

VP9 results are directly comparable only when both snapshots carry QpMode VP9 Base Q Index. AV1 and VP9 Base Q modes remain distinct even though both use a 0 through 255 index domain.

11. Validation and Defensive Parsing

11.1 Boundary checks

- Canonical samples require non-negative offsets and positive lengths.
- Offset plus length must remain within the source file.
- VP9 sample lengths must fit the in-memory parser contract.
- A superframe index must fit inside the sample, repeat its marker, contain positive frame sizes, and account exactly for the payload before the index.
- Every internal frame reader is bounded to the indexed frame length.
- Frame marker, profile bits, sync code, RGB profile constraints, reference-slot availability, and Base Q range are validated.

11.2 Error model

A VP9 syntax failure increments Headers rejected, preserves the first failure text, marks the sample frame type as Other, and prevents histogram activation. No guessed quantizer value is substituted.

11.3 Runtime isolation

- No external multimedia executable is launched.
- No temporary media file is created.
- The source is opened read-only with shared read access.
- The single-file script remains compatible with in-memory EXE hosting.

12. Supported Combinations

Video	Matroska/WebM	MP4/M4V/MOV	AVI
H.264 / AVC	Yes	Yes	Yes, Annex B
H.265 / HEVC	Yes	Yes	No
AV1	Yes	Yes	No
VP9	Yes, validated	Mapped through canonical pipeline	No
MPEG-4 Part 2 / XviD	Yes	Yes	Yes

12.1 VP9 ISO Base Media route

- Recognized sample entries: vp09 and legacy vp08.
- Native codec identity: V_VP9.
- Canonical family and payload: VP9 / Vp9Frame.
- Optional configuration record: vpcC.
- Supported container structure: standard non-fragmented moov-based sample tables.
- Fragmented MP4 using moof/traf/trun remains unsupported.

12.2 Comparison rule

Quantizer comparison requires matching QpMode. VP9 Base Q Index is not overlaid with AV1 Base Q Index.

13. Known Limitations

- Fragmented MP4 is not supported. MP4/MOV support is limited to non-fragmented files with conventional sample tables in moov.
- VP9 is regression validated for Matroska/WebM V_VP9 and for the vp09 sample entry in standard non-fragmented MP4/M4V/MOV, using Profile 0, 8-bit YUV 4:2:0 assets. Legacy vp08 mapping is implemented but has not been separately runtime tested.
- VP9 in AVI is not enabled.
- The VP9 histogram is frame-level Base Q Index, not a reconstructed per-block distribution after segmentation.
- The UI supports one Current A and one Reference B, not batch ranking.
- LSD does not decode pixels and does not calculate PSNR, SSIM, VMAF, or related image metrics.
- AVI AAC and AVI E-AC-3 mappings are not enabled.
- Container-only color metadata is not used as a fallback when codec signaling is absent.
- Bitrate differences across unlike resolution, frame rate, or duration are factual measurements, not a quality verdict.

14. Verification and Regression Matrix

Asset	Expected VP9 result
10-second Matroska sample	300 samples; 325 internal frames; 25 superframe samples; 2 KEY; 323 INTER; Q 31-179; zero rejected
Full Matroska file	18,977 samples; 19,176 internal frames; 199 superframe samples; 135 KEY; 19,041 INTER; Q 0-179; zero rejected

14.1 MP4 mapping verification

- Static route: vp09/vp08 -> V_VP9 -> CodecFamily VP9 -> PayloadFormat Vp9Frame -> ParseVp9Payload.
- Optional vpcC lookup must remain parent-bounded.
- Canonical sample count must equal the resolved MP4 sample-table count.
- The validated vp09 regression asset contains 300 video samples, 300 parsed VP9 frames, 2 KEY frames, 298 INTER frames, zero rejected headers, and Base Q Index range 46-190.
- Packet/frame accounting, shown/sample consistency, Base Q completeness, and VP9 validation all pass.

14.2 Required invariants

- $\text{InternalFrames} = \text{Parsed} + \text{ShowExisting}$.
- $\text{Parsed} = \text{KeyFrames} + \text{InterFrames}$ for newly coded frames.
- $\text{Histogram count} = \text{Parsed}$ when $\text{Rejected} = 0$.
- $\text{Displayed I} + \text{P} + \text{B} + \text{Other} = \text{container video sample count}$.

15. Maintenance Guidelines

15.1 MP4 VP9 extension rules

- Keep mapping inside the existing stsd video sample-entry branch.
- Do not introduce a second MP4-specific VP9 frame parser.
- Retain optional vpcC bytes without requiring them for payload traversal.
- Reuse existing sample tables, canonical bounded samples, diagnostics, summary block, and histogram.
- Retain the deterministic vp09 MP4 regression asset and its expected counters in future releases.

15.2 Code that should remain stable

- Canonical model contracts and sample boundary validation.
- Existing Matroska VP9 regression behavior.
- AVI/OpenDML chunk resolution and MP4 parent-bound checks.
- A/B role semantics and snapshot immutability.
- N/A reporting and rejected-unit visibility.

15.3 Future candidates

- Fragmented MP4 sample indexing.
- VP9 Profiles 1, 2, and 3 regression assets.
- Container-color fallback through MP4 colr/nclx and Matroska Colour.
- Additional video codec families.
- Optional batch analysis and textual A/B compatibility summary.

Appendix A. Key Types and Responsibilities

Type / state	Responsibility
NativeMediaInfo / NativeTrack	Container-native metadata and codec configuration
LsdMediaModel / LsdTrackModel / LsdSampleModel	Container-neutral tracks and bounded samples
NativeMediaProbe / NativeMp4Probe / NativeAviProbe	Container detection, metadata, and sample indexing
NativeMatroskaPacketScan	Shared canonical video analysis job; historical type name
Vp9BitReader	Bounded MSB-first VP9 bit access
Vp9FrameParser	Superframe split, frame-header traversal, reference state, Base Q extraction
Vp9FrameInfo / Vp9ChunkInfo	Per-frame syntax result and per-sample internal-frame collection
CanonicalAudioAnalyzer	MP3, Dolby, and PCM validation/accounting
ReferenceResult	Frozen B file, bitrate bins, QP mode/counts, and summary
PowerShell coordinator	Jobs, state, reports, graphs, comparison, and GUI

Appendix B. Terminology

Term	Definition
Canonical sample	Container-neutral record describing a bounded payload range and timing
Internal VP9 frame	One coded VP9 frame extracted directly from a sample or its superframe payload
VP9 superframe	One sample containing multiple concatenated coded frames and a trailing size index
Shown frame event	Presentation caused by a newly coded shown frame or show_existing_frame
Hidden coded frame	A newly coded frame with show_frame equal to zero
show_existing_frame	Display an existing reference slot without coding a new frame or Base Q Index
KEY_FRAME	VP9 intra frame with key-frame semantics and full reference refresh
INTER_FRAME	VP9 inter-coded frame; hidden alternate-reference frames are included
Base Q Index	Frame-level native quantizer lookup index, 0 through 255 for AV1 and VP9
Current A	Most recently analyzed active result, rendered blue
Reference B	Stored comparison result, rendered gold

Appendix C. Release Identification

Property	Value
Product	Little Stream Detector
Version	3.0
Documentation edition	VP9 and standard MP4/MOV mapping
Runtime model	Single PowerShell file with embedded C#
Runtime dependencies	Windows PowerShell 5.1 and .NET Framework classes
External multimedia tools	None
Temporary media files	None
Validated VP9 route	Matroska/WebM V_VP9 and non-fragmented MP4 vp09, Profile 0, 8-bit YUV 4:2:0
Implemented VP9 route	Legacy non-fragmented MP4 vp08 mapping; separate runtime test pending
Unsupported route	Fragmented MP4 and VP9 in AVI

Release status. LSD 3.0 retains the validated Matroska/WebM VP9 parser and routes validated ISO Base Media vp09 samples to the same canonical analysis path. Legacy vp08 mapping is also implemented. The optional vpcC box is captured as codec private data. The codebase remains a single native PowerShell application with embedded C# and no external multimedia dependency.